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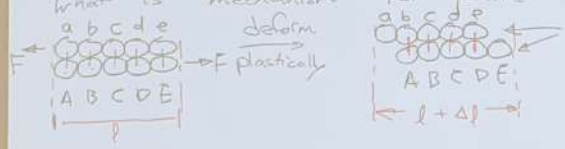
Charles Jia

True Stress in neck continues to increase (true stress)

(aside $\sigma_T = \frac{F}{A_i}$ ← instantaneous cross-sectional Area)

Plastic deformation, dislocations, defects and strengthening mechanisms

What is mechanism for plastic deformation? (Perfect crystal)



deform plastically

close-packed planes sliding

bond breaking + bond re-forming

(aA, bB, cC, dD, eE) → (bA, cB, dC, eD)

Metals (real crystals are full of defects or imperfections

- dislocation (Extra half plane) edge dislocation

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what is mechanism for plastic deformation? (Perfect crystal)

a b c d e — F — A B C D E — l

deform plastically → F

a b c d e — A B C D E — l + Δl

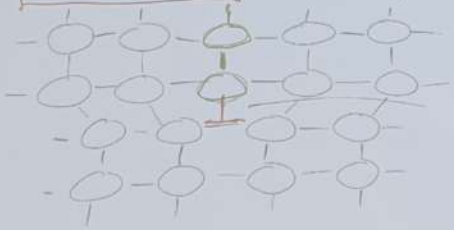
close-packed planes | sliding

bond breaking + bond re-forming

(aA, bB, cC, dD, eE) → (bA, cB, dC, eD)

Metals (real crystals are full of | defects | or | imperfections |

- | dislocation | (Extra half plane) edge dislocation



What actually happens (not a model this time, b/c evidence is so strong)

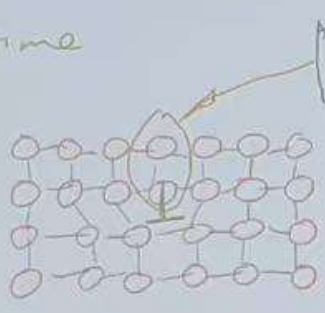
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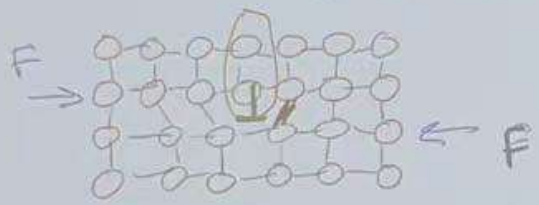


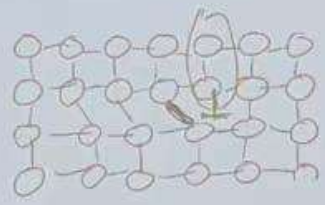
What actually happens (not a model this time, b/c evidence is so strong) is that bonds break and reform one row of atoms at a time

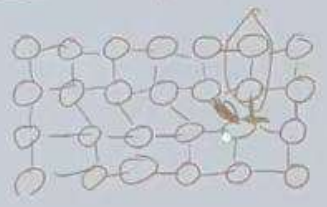


metals are full of imperfections like these!

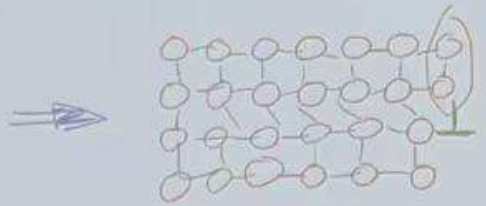
Dislocations. ⊥ symbol







I





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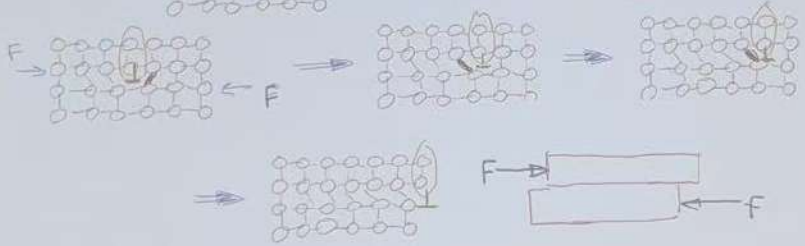
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at a time

extra half-plane

metals are full of imperfections like those!

Dislocations. ⊥ symbol



I

☆ plastic deformation is due to crystal plane sliding facilitated by dislocation movement

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☆ plastic deformation is due to crystal plane sliding facilitated by dislocation movement

Oct. 7 2026

- How to strengthen a meta ?
- Inhibiting dislocation movement hinders plastic deformation and strengthens a metal. ($\sigma_y \uparrow$)

Creating defects/imperfects

1-D defect : dislocation
edge

- The edge of the extra half-plane is a line (dislocation line)

So, dislocations are 1-D defects

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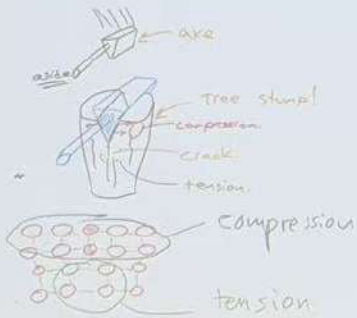
deformation and strengthens a metal. ($\sigma_y \uparrow$)

Creating defects/imperfects

1-D defect: ~~dislocation~~ edge dislocation

- The edge of the extra half-plane is a line (dislocation line)

So, dislocations are 1-D defects



Defects → elastic lattice distortion

Spatial distribution of distorted lattice forms strain field or lattice strain.

Strain → stress → strain energy in strain field

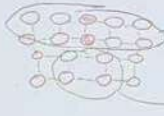
} strain field/strain energy

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Spatial distribution of distorted lattice forms strain field or lattice strain.

Strain → stress → strain energy in strain field



} strain field / strain energy

Crystalline imperfections.

But please remember that these imperfections also change other properties, like electrical, optical, corrosive etc

→ Lets organize based on dimensionality

0D - missing atoms, extra atoms, different atoms

1D - Dislocations

2D - Interfaces; surfaces (grain boundary)

3D - things with volume.

Let's make steel: Fe + C add carbon to iron (0D inorganic Imperfection) or Point Defect



R_{Fe} ~ 140 - 160 pm } so carbon dissolves

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Let's make steel add carbon to iron (0D inversional Imperfection) or Point Defect



$R_{Fe} \sim 140-160$ pm

$R_C \sim 70$ pm

} so carbon dissolves into interstitial sites.

0D – interstitial point defect

a)

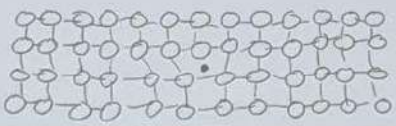


But there is already lattice strain around a dislocation

Which has a lower strain energy?

b)

so which is more stable?



lower energy

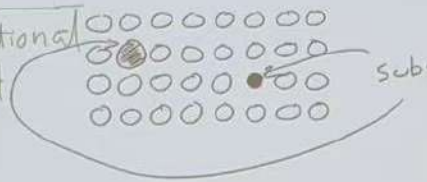
Carbon strengthens iron,

hindering dislocation movement

impurities act to pin dislocations.

also can have substitutional impurities:

0D – substitutional point defect

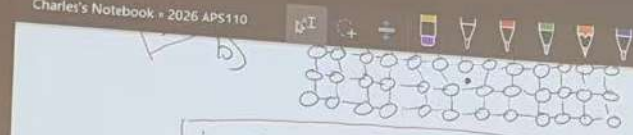


normally within ~10% radii, same crystal structure, similar electronegativity.

Alloying $Cu + Zn$ → brass

• One more 0D imperfection that is very important,

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Impurities act to pin dislocations.

hindering dislocation movement

also can have substitutional impurities:

0D - substitutional point defect

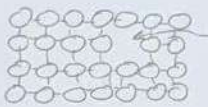


substitutional

normally within ~10% radii, same crystal structure, similar electronegativity

Alloying Cu + Zn → brass

- One more 0D imperfection that is very important, but not so much in mechanical properties is the vacancy



vacancy

This is the result of an atom occasionally making a successful jump out of it's lattice site

3 0D - interstitial (extra atoms), substitutional (different atom)
- vacancy (missing atom)

- 2-Dimensional Imperfections

- Free surfaces
  - unsatisfied bonds (fewer nearest

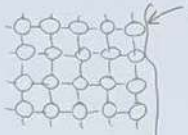


free surface

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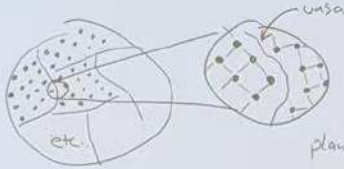
2-Dimensional Imperfections

- Free surfaces
  - unsatisfied bonds (fewer nearest neighbours)



elevated energy = surface energy

Internal surfaces (Grain boundaries)
- similar to free surfaces.



Consider a dislocation moving from one grain to another ⇒ must
- change direction
- navigate planar mismatch
- overcome some lattice strain

planar mismatch

inhibits dislocation movement

So, which will be stronger?



Smaller grains = stronger

"Grain size reduction"

3 Dimensional
- Pores

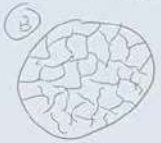
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one grain to another ⇒ must: · change direction
· navigate phase mismatch
· overcome some lattice strain

mismatch
inhibits dislocation movement

So, which will be stronger?





Smaller grains = stronger
"Grain Size reduction"

3 Dimensional
- Pores
- 2nd phase particles: Fe_3C in steel · normally hard, brittle phases

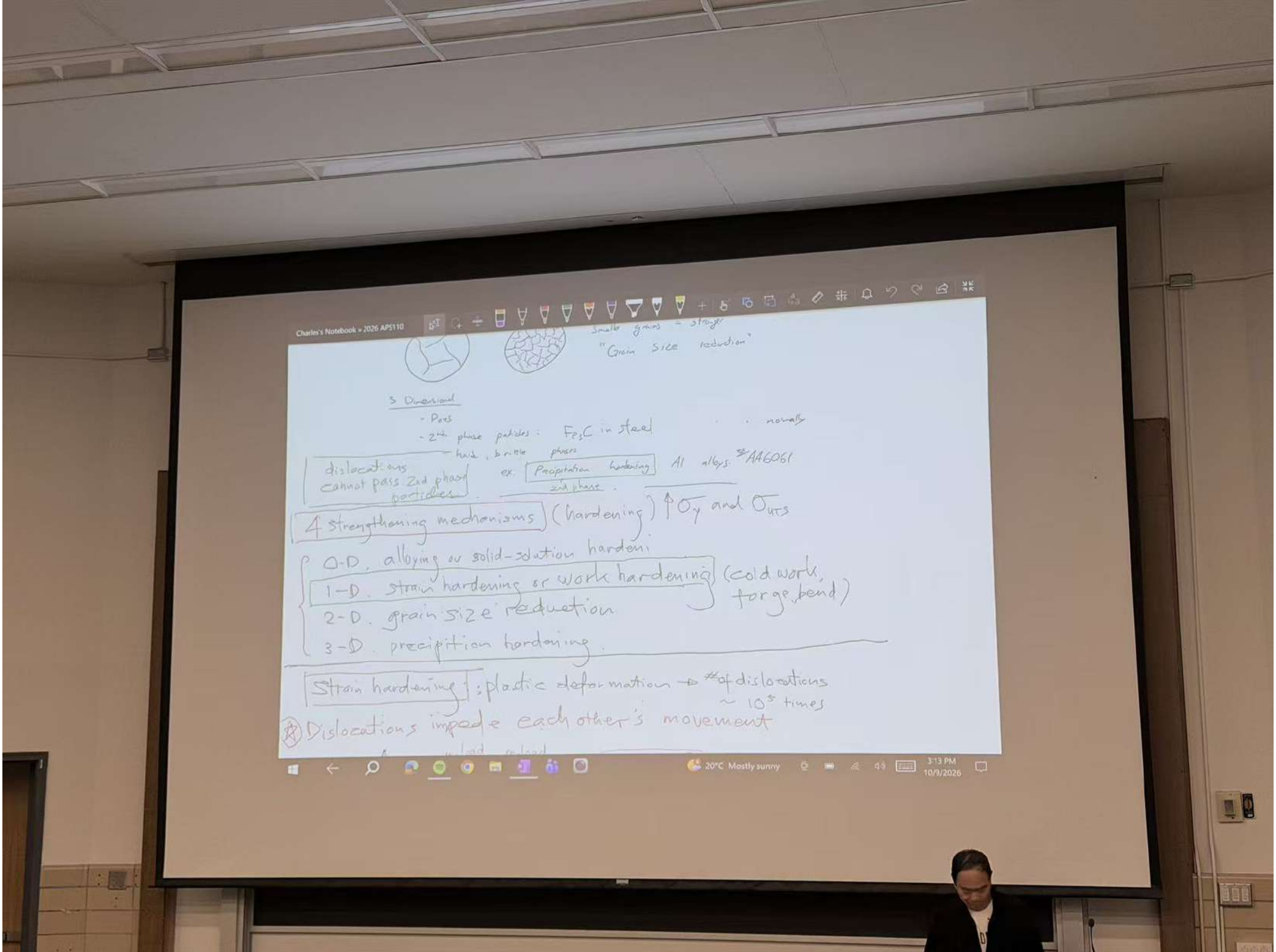
dislocations cannot pass 2nd phase particles

ex. Precipitation hardening Al alloys. ex AA6061
2nd phase

strengthening mechanisms (hardening)

0-D, alloying or solid-solution hardening

1-D strain hardening (cold working)

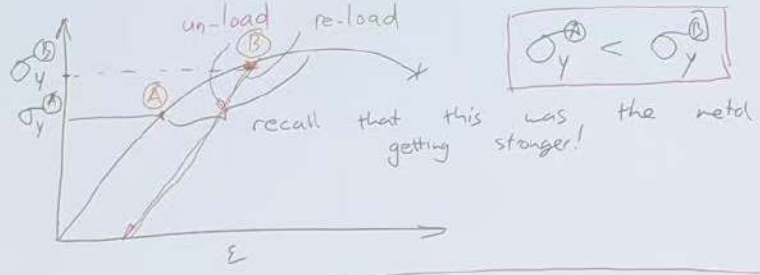

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smaller grains = stronger
"Grain size reduction"
3 Dimensional
- Pores
- 2nd phase particles: Fe_3C in steel
hard, brittle phases
dislocations cannot pass 2nd phase particles
ex. Precipitation hardening
Al alloys. ex. AA6061
2nd phase
4 strengthening mechanisms (hardening) ↑ σ_y and σ_{UTS}
0-D. alloying or solid-solution hardening
1-D. strain hardening or work hardening (cold work, forge, bend)
2-D. grain size reduction
3-D. precipitation hardening
Strain hardening: plastic deformation → # of dislocations ~ 10^5 times
Dislocations impede each other's movement
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(5-1). precipitation hardening.

Strain hardening: plastic deformation → # of dislocations ~ 10^5 times

(4) Dislocations impede each other's movement

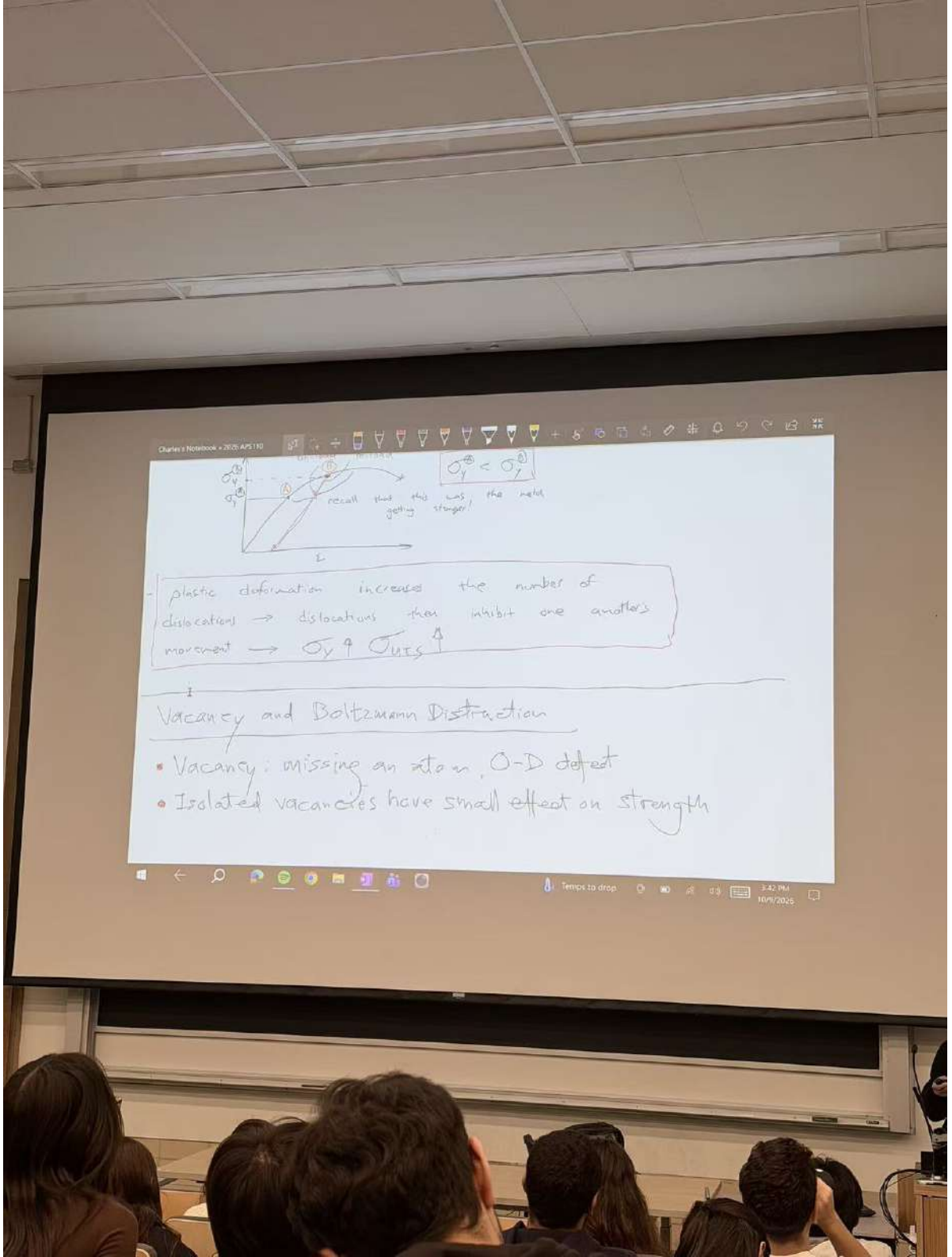


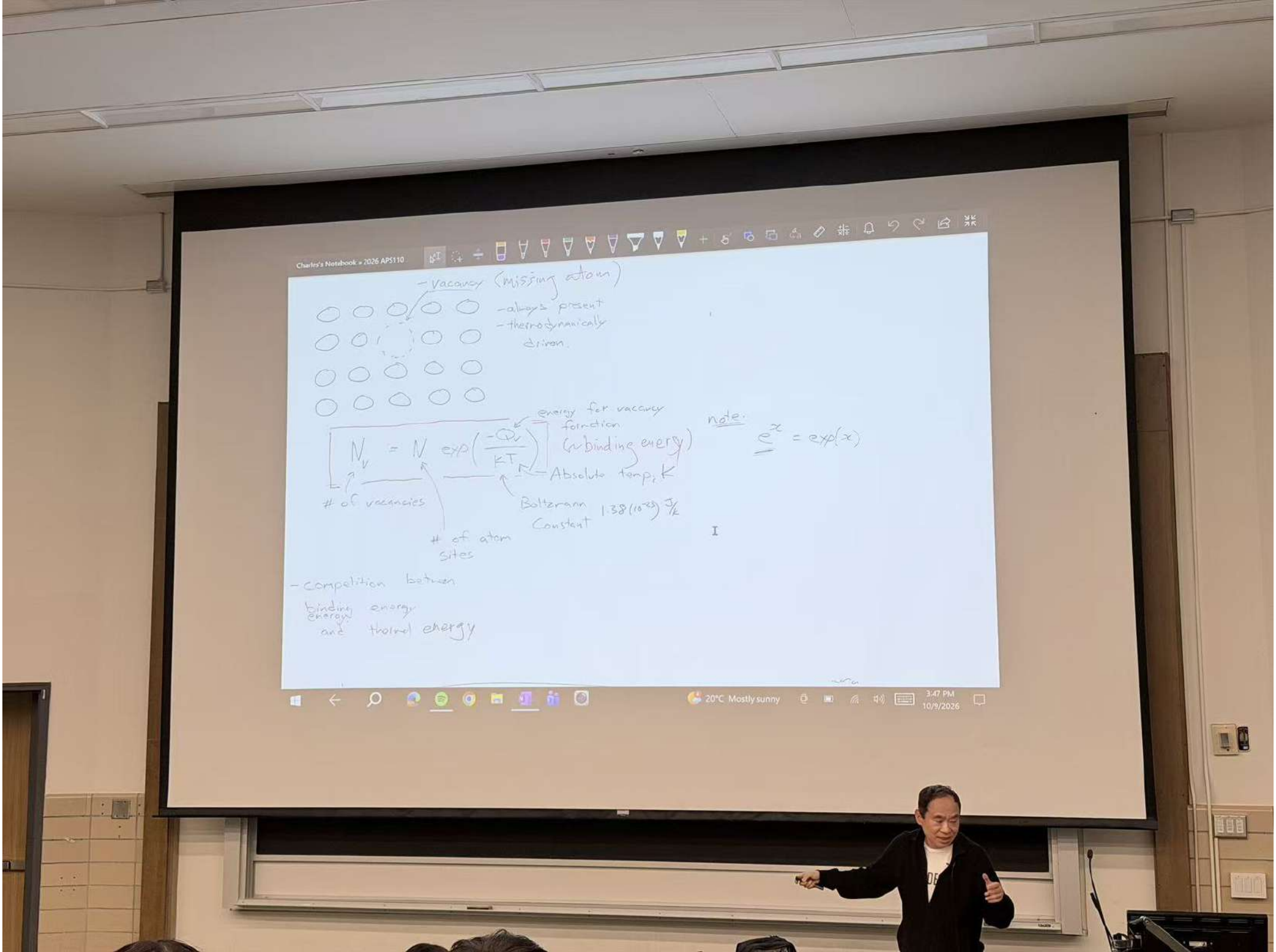
$\sigma_y^A < \sigma_y^B$

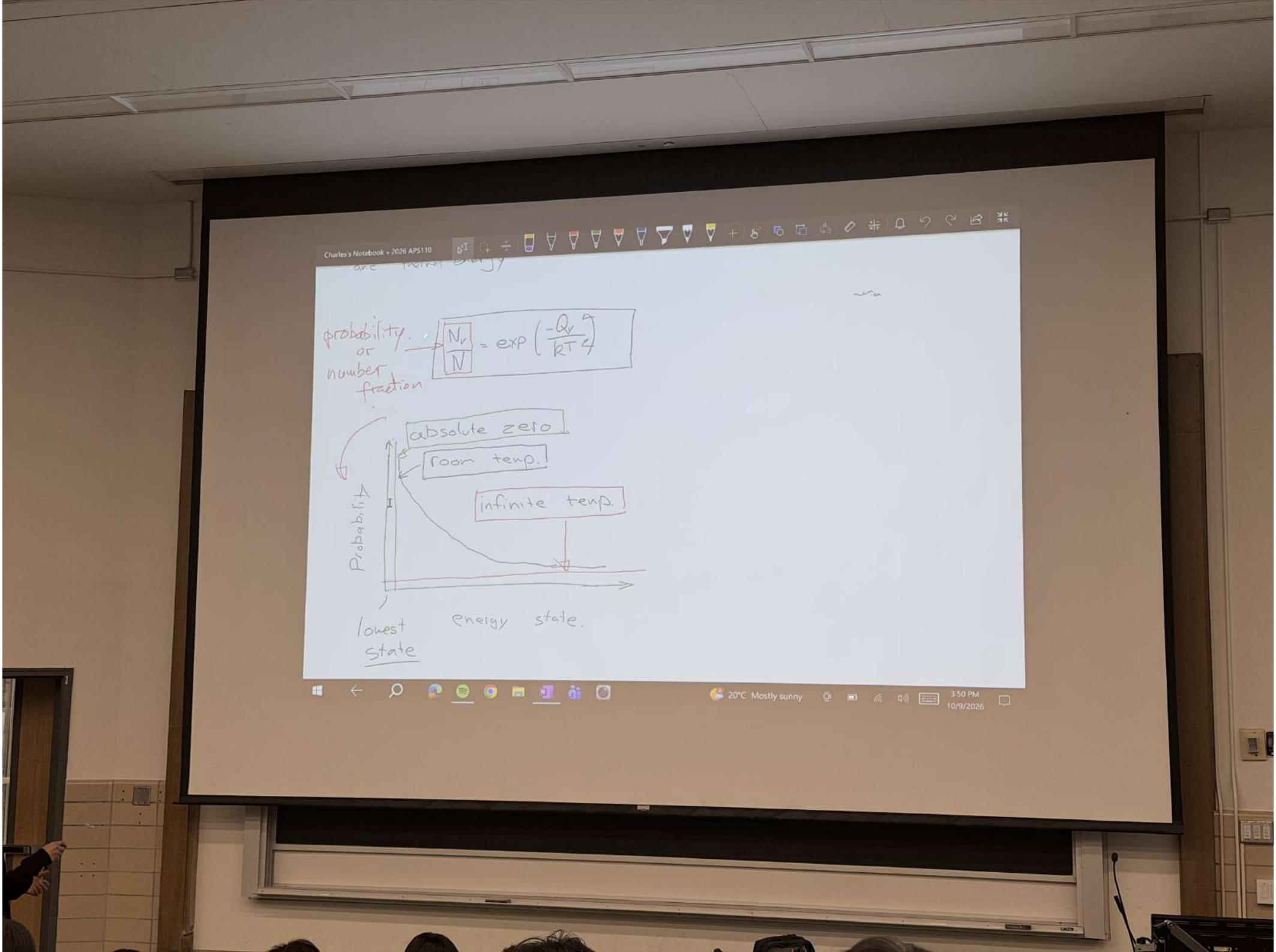
plastic deformation increases the number of dislocations → dislocations then inhibit one another's movement → $\sigma_y \uparrow$ $\sigma_{UTS} \uparrow$

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recall that getting this was stronger! the metal
plastic deformation increases the number of dislocations → dislocations then inhibit one another's movement → σy ↑ σUTS ↑
Vacancy and Boltzmann Distribution
Vacancy: missing an atom, 0-D defect
Isolated vacancies have small effect on strength
Temps to drop
3:42 PM
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- vacancy (missing atom)
- always present
- thermodynamically driven.
energy for vacancy formation (or binding energy)
$N_v = N \exp\left(\frac{-Q_v}{kT}\right)$
of vacancies
of atom sites
Absolute temp, K
Boltzmann Constant $1.38(10^{-23})$ J/K
note: $e^x = \exp(x)$
- competition between binding energy and thermal energy
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probability or number fraction
$\frac{N_v}{N} = \exp\left(\frac{-Q_v}{kT}\right)$
absolute zero
room temp.
infinite temp.
Probability
lowest energy state.
state
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